SDVD / Valid Dictionary (VD): First Draft Notes

1 Motivation and scope

These notes are a first draft of a proposal for formalising how scientific language is built and used. The core claim is not that we have discovered the "true meanings" of words, but that we can write down a disciplined *structure* in which statements about words, definitions, and laws are made explicit and therefore debateable, comparable, and repairable.

The proposed framework is the *Valid Dictionary* (VD), implemented as a named dictionary authority. A concrete instance used throughout is the *Super Duper Valid Dictionary* (SDVD), a deliberately unobtrusive name intended to avoid premature sociological escalation ("this is the one true dictionary of science") while still enabling precise reference.

2 VD as a named data structure

2.1 Dictionary object and the necessity of a name

At the most primitive level, a dictionary is a data structure:

$$D : \text{Strings} \to \text{Strings},$$

mapping an input string (a "word") to an output string (its dictionary entry).

A subtle but foundational point is that, in practice, one cannot speak about "the definition of a word" without implicitly relying on an *authority*—some specific dictionary or convention. The VD proposal makes that authority explicit by treating the dictionary as a *named object*.

Definition 1 (Named dictionary authority)**.** *A* named dictionary *is a pair* $(\text{Name}(D), D)$ *where* $\text{Name}(D) \in \text{Strings}$ *and* $D : \text{Strings} \to \text{Strings}$.

Remark 1 (Authority indexing)**.** *Statements of the form "the definition of w is δ" should be read as*

$$\textit{"In dictionary } D\textit{, } D(w) = \delta\textit{."}$$

Without the dictionary name (explicit or implicit), the statement is ill-posed.

2.2 Democratising authority

Once authority is explicit, it is also *contestable and replicable*. One may propose SDVD, an alternative VD_2, or any other named dictionary, and compare them as objects. This turns many philosophical disputes into explicit disagreements about dictionary entries and their induced behaviour, rather than disputes conducted under a hidden background authority.

3 The first three SDVD entries

We now install three core entries in SDVD. They act as type-setting primitives: they specify what sort of objects "definitions", "meanings", and "physical laws" are taken to be *in this framework*.

3.1 Definition: definition is the dictionary output

Definition 2 (Definition in SDVD (DoD))**.** *In SDVD, a* definition *is the output string returned by the named dictionary when queried with a word. Two raw definitions are different if even one character differs.*

This is intentionally syntactic. It separates *dictionary content* (strings) from whatever mental, physical, or social phenomena those strings may trigger.

3.2 Meaning: meaning is what an utterance does

A dictionary entry is a string; meaning is not required to be a string. The VD proposal treats meaning as a causal notion: the effect of an utterance on the world.

Definition 3 (Meaning in SDVD (DoM))**.** *In SDVD, the* meaning *of a word is the total change in the relevant state of the world (physical, cognitive, computational, etc.) caused by the utterance of that word in a given context.*

Remark 2 (Wavefunction as an instantiation)**.** *In quantum-mechanical settings, the "relevant state" may be represented as a wavefunction; in other settings it may be a belief state, internal robot registers, a social/behavioural state, or even the acoustic field. The definition is intended to be* model-agnostic*: choose the state representation appropriate to the discussion.*

3.3 Physical law: a triplet entry (structural definition)

The definition of physical law is kept purely structural: it describes a pattern of dictionary introduction.

Definition 4 (Physical law in SDVD)**.** *A* physical law *is the* ***first entry*** *in a* ***triplet*** *of dictionary entries in which:*

(i) the first word is introduced together with (and explicitly evokes) two previously undefined terms, and

(ii) the three terms are mutually locked*: none can be fully defined without reference to the other two.*

This definition intentionally does not attempt to encode truth, experiment, or utility. Those topics may be discussed as external selection pressures on which triplets are retained by a community, but they are not built into the dictionary-level definition of "physical law".

4 The Pundit Square: tracking definition/meaning interactions

Words like "definition" and "meaning" are themselves words whose behaviour must be tracked. A compact bookkeeping device is the *Pundit Square*, a 2×2 grid:

Definition of Definition (DoD)	**Definition of Meaning (DoM)**
Meaning of Definition (MoD)	**Meaning of Meaning (MoM)**

4.1 Pedagogical order

A practical point: many readers interpret "definition" as "mathematical characterisation" by default. To avoid this confusion, a good learning order is:

$$\textbf{DoD} \to \textbf{DoM} \to \textbf{MoD} \to \textbf{MoM}.$$

First nail down the syntactic dictionary notion of definition; then ground meaning as causal effect; only then explain how the word "definition" behaves in human/mathematical practice; finally explain the meta-behaviour of the word "meaning".

5 DoM with examples: meaning as causal impact

Under DoM, an utterance can have many types of effects:

- **Physical:** air vibrations, recordings, signals in wires, actuation on sensors.
- **Cognitive:** recalling a stored definition, shifting attention, changing expectations.
- **Behavioural/social:** changing decisions, eliciting actions, coordinating groups.
- **Computational:** updating internal registers, running a procedure, logging an event.

Thus meaning is not reduced to a private mental "feeling" nor to a static semantic object; it is the world-state update induced by an utterance.

A key interaction with the dictionary is this: uttering a word can increase the probability that listeners explicitly realise (or "load into working memory") particular definition strings stored in their own internal dictionaries.

6 MoD: mathematical equality vs raw definitional identity

A recurring confusion is that the word "definition" often means something like "characterisation up to mathematical equality". VD makes it possible to separate these notions cleanly.

6.1 Raw definitional identity

By construction, raw definitional identity is character-level identity of strings. If δ_1 and δ_2 differ by one character, they are distinct raw definitions.

6.2 Mathematical equality is a different relation

In ordinary mathematics one writes

$$4 = 2 + 2 = 1 + 3.$$

This is not a claim that the strings "4", "2+2", and "1+3" are identical definitions; they are not. Instead the equality sign "=" is carrying a *mathematical equality* relation, e.g. equality of value in $(\mathbb{N}, +)$.

Remark 3 (VD is compatible with mathematics)**.** *VD does not reject mathematical language; it clarifies that mathematical "=" is not raw string identity. Mathematics is an arena where both definitions and the operational meaning of expressions are unusually tightly policed.*

7 MoM: why language works, why debates break, and how VD helps

The *meaning of meaning* is the recognition that (i) definitions are private and variable, while (ii) communication succeeds because the *causal effects* of utterances (meanings) overlap sufficiently across agents.

7.1 Different definitions, aligned meanings

Two people almost never store identical definition strings for a word; yet they can coordinate because utterances push them into similar enough predictive/behavioural states. Communication is therefore primarily supported by overlap of meanings, not identity of raw definitions.

7.2 Why natural language is easy and mathematics is hard

Natural language tolerates large definitional variation because approximate meaning-overlap often suffices in context. Mathematics is difficult precisely because it demands tighter policing: both definitions and the operational meaning of expressions (procedures performed by people and machines) must align sharply to support long inference chains.

7.3 Why apparent agreement collapses in debate

Casual agreement can be an illusion: agents may share enough meaning-overlap for everyday coordination while harbouring divergent definitions that only become visible under pressure (debate, edge cases, formal derivation). VD diagnoses this as a mismatch between definition alignment and meaning alignment.

7.4 Yielding authority to SDVD as a repair protocol

If disputants yield authority to a shared named dictionary (e.g. SDVD), they gain a protocol:

(i) index claims to explicit entries (stop smuggling hidden authorities),

(ii) separate definitional disagreement (dictionary edits/splits) from empirical disagreement (world facts),

(iii) split overloaded symbols/words into distinct entries when needed,

(iv) continue discussion under explicit rules despite language slipperiness.

8 Triplets and physical law structure

8.1 Abstract triplet pattern

Let D be a named dictionary. A triplet involves three words (w_0, w_1, w_2) and three entries $D(w_i)$ such that:

(i) w_0 is introduced by an entry that explicitly invokes two previously undefined terms w_1, w_2,

(ii) $D(w_1)$ references w_0 and w_2,

(iii) $D(w_2)$ references w_0 and w_1,

so that the dependency graph forms a 3-cycle. This is the minimal closure structure preventing infinite definitional regress.

8.2 Triplets can chain

Large mathematical statements can be decomposed into networks of triplets (by introducing intermediate names for subexpressions). In practice one often keeps equations compact to avoid naming explosion, while recognising that a triplet decomposition exists in principle.

9 Laws, operations, and "meaningful relationships"

A physical or mathematical law is not merely a list of definitions; it also specifies operations and relations that tell a capable agent what to do with numbers or procedures.

Even in arithmetic, defining an addition operator does not by itself make the statement "$2 + 2 = 4$" hold as a raw definitional identity. Rather, the equation asserts that when a capable system engages the relevant computation under the accepted rules, the values coincide. This

illustrates a general point: the non-term symbols in laws ("+", "=", "proportional to", etc.) carry binding operational meaning.

(1) Symbol overloading and dictionary collisions. Physics notation often reuses the same glyph for distinct quantities (e.g. writing F in both $F = ma$ and $F = kx$), relying on human intuition and context to disambiguate. In VD terms this is a *dictionary collision*: multiple concepts are implicitly mapped onto the same symbol-entry, and meaning is patched in "by vibe" rather than by explicit definition. A VD-aligned discipline therefore prefers naming hygiene (e.g. F_{net}, F_T, F_g) and explicit splits of overloaded symbols, because otherwise definitional drift is easy, learning becomes harder, and even capable reasoning systems can silently conflate distinct entries.

(2) "Definition" in ordinary practice vs. SDVD "definition". Many readers hear the word *definition* and import a mathematical notion ("characterisation up to mathematical equality") or an essentialist notion ("the true nature of the thing"). SDVD deliberately starts with a narrower syntactic primitive: *definition* means the literal output string of a named dictionary, with identity determined character-by-character. The pedagogical order DoD→DoM→MoD→MoM is designed to prevent confusion: one first fixes the string-level object, then grounds meaning as causal impact, and only then explains how mathematical and everyday uses of "definition" relate to (and often overload) the SDVD usage.

(3) Equality is overloaded: raw identity vs. mathematical equality. The glyph "=" carries multiple distinct relations across domains. Under raw_definition, equality is *string identity*: two definitions are equal only if their character strings are identical. In mathematics, however, "=" typically denotes a *value-equality* relation inside a structure (e.g. equality of denotation in $(\mathbb{N}, +)$), allowing different strings such as "$2 + 2$" and "$1 + 3$" to be equal even though they are not identical as strings. VD is compatible with mathematics precisely because it forces this disambiguation: whenever "=" is used, one must make explicit which equality relation is intended rather than silently sliding between them.

10 Outlook (not yet finalised)

The present notes motivate several necessary extensions:

- **Governance rules:** how laws are added, revised, restricted in scope, or removed.
- **Question-answering:** how "why?" chains traverse triplets and where closure is obtained.
- **Causality:** how causal language emerges from update rules, interventions, and directed traversals.
- **Stopping regress:** formalising why repeated "why?" loops around triplets unless anchored by external procedures.